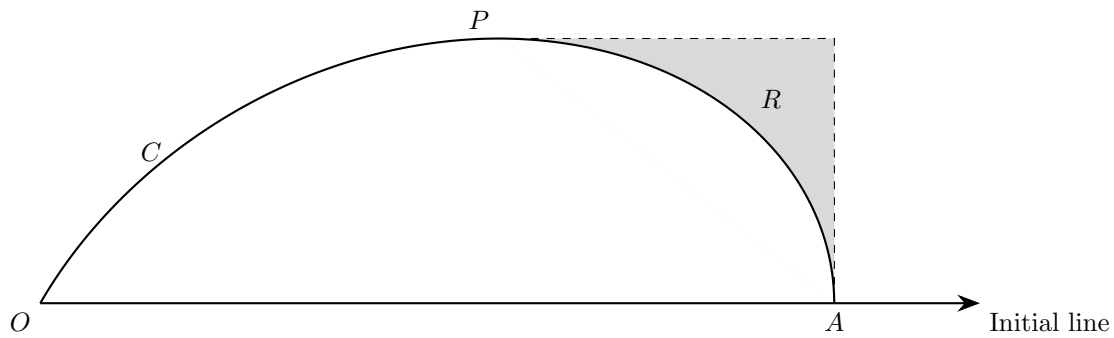


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1. The curve C shown in the figure above has polar equation

$$r = 1 + 2 \cos 2\theta, \quad 0 \leq \theta \leq \frac{\pi}{3}$$

At the point P on C , the tangent to C is parallel to the initial line.

- (a) Show that P has polar coordinates $\left(2, \frac{\pi}{6}\right)$. [4]

The point A on C has polar coordinates $(3, 0)$.

The finite region R , shown shaded in the figure above, is bounded by the arc AP of C , the tangent to C at P and the line through A parallel to $\theta = \frac{\pi}{2}$.

- (b) Find the exact area of R . [9]

Solution

- (a) Using

$$x = r \cos \theta, \quad y = r \sin \theta$$

with

$$r = 1 + 2 \cos 2\theta$$

we have

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

Since the tangent is parallel to the initial line, it is horizontal, so

$$\frac{dy}{d\theta} = 0 \quad \text{and} \quad \frac{dx}{d\theta} \neq 0$$

First,

$$\frac{dr}{d\theta} = -4 \sin 2\theta$$

Now

$$\begin{aligned} \frac{dy}{d\theta} &= \frac{d}{d\theta}(r \sin \theta) \\ &= \frac{dr}{d\theta} \sin \theta + r \cos \theta \\ &= -4 \sin 2\theta \sin \theta + (1 + 2 \cos 2\theta) \cos \theta \\ &= -8 \sin^2 \theta \cos \theta + (1 + 2(1 - 2 \sin^2 \theta)) \cos \theta \\ &= -8 \sin^2 \theta \cos \theta + (3 - 4 \sin^2 \theta) \cos \theta \\ &= 3 \cos \theta (1 - 4 \sin^2 \theta) \end{aligned}$$

So

$$3 \cos \theta (1 - 4 \sin^2 \theta) = 0$$

For $0 \leq \theta \leq \frac{\pi}{3}$, we have $\cos \theta > 0$, so

$$1 - 4 \sin^2 \theta = 0$$

Hence

$$\sin \theta = \frac{1}{2}$$

and in the given interval this gives

$$\theta = \frac{\pi}{6}$$

Now find r :

$$r = 1 + 2 \cos \left(2 \cdot \frac{\pi}{6} \right) = 1 + 2 \cos \frac{\pi}{3} = 1 + 2 \cdot \frac{1}{2} = 2$$

Also,

$$\frac{dx}{d\theta} = \frac{d}{d\theta}(r \cos \theta) = \frac{dr}{d\theta} \cos \theta - r \sin \theta$$

and at $\theta = \frac{\pi}{6}$,

$$\frac{dx}{d\theta} = (-2\sqrt{3}) \cdot \frac{\sqrt{3}}{2} - 2 \cdot \frac{1}{2} = -3 - 1 = -4 \neq 0$$

so this does give a horizontal tangent.

Therefore P has polar coordinates

$$\left(2, \frac{\pi}{6} \right)$$

(b) Let Q be the point where the tangent at P meets the vertical line through A .

From part (a),

$$P = \left(2, \frac{\pi}{6} \right)$$

so in Cartesian coordinates,

$$P = \left(2 \cos \frac{\pi}{6}, 2 \sin \frac{\pi}{6} \right) = (\sqrt{3}, 1)$$

Also $A = (3, 0)$, since its polar coordinates are $(3, 0)$.

The tangent at P is parallel to the initial line, so it is

$$y = 1$$

The line through A parallel to $\theta = \frac{\pi}{2}$ is vertical, so it is

$$x = 3$$

Hence

$$Q = (3, 1)$$

If the arc AP were replaced by the chord AP , the region would be triangle AQP . So

$$\text{Area}(R) = \text{Area}(\triangle AQP) - \text{area between chord } AP \text{ and arc } AP$$

First,

$$AQ = 1, \quad QP = 3 - \sqrt{3}$$

so

$$\text{Area}(\triangle AQP) = \frac{1}{2}(1)(3 - \sqrt{3}) = \frac{3 - \sqrt{3}}{2}$$

Now find the area between the chord and the arc.

This is

$$\text{Area of sector } OAP - \text{Area}(\triangle OAP)$$

For the sector,

$$\text{Area of sector } OAP = \frac{1}{2} \int_0^{\pi/6} r^2 d\theta = \frac{1}{2} \int_0^{\pi/6} (1 + 2 \cos 2\theta)^2 d\theta$$

Expand the square:

$$\begin{aligned} (1 + 2 \cos 2\theta)^2 &= 1 + 4 \cos 2\theta + 4 \cos^2 2\theta \\ &= 1 + 4 \cos 2\theta + 2(1 + \cos 4\theta) \\ &= 3 + 4 \cos 2\theta + 2 \cos 4\theta \end{aligned}$$

So

$$\begin{aligned} \text{Area of sector } OAP &= \frac{1}{2} \int_0^{\pi/6} (3 + 4 \cos 2\theta + 2 \cos 4\theta) d\theta \\ &= \frac{1}{2} \left[3\theta + 2 \sin 2\theta + \frac{1}{2} \sin 4\theta \right]_0^{\pi/6} \\ &= \frac{1}{2} \left(\frac{\pi}{2} + 2 \sin \frac{\pi}{3} + \frac{1}{2} \sin \frac{2\pi}{3} \right) \\ &= \frac{1}{2} \left(\frac{\pi}{2} + 2 \cdot \frac{\sqrt{3}}{2} + \frac{1}{2} \cdot \frac{\sqrt{3}}{2} \right) \\ &= \frac{1}{2} \left(\frac{\pi}{2} + \sqrt{3} + \frac{\sqrt{3}}{4} \right) \\ &= \frac{\pi}{4} + \frac{5\sqrt{3}}{8} \end{aligned}$$

For triangle OAP ,

$$OA = 3, \quad OP = 2, \quad \angle AOP = \frac{\pi}{6}$$

so

$$\text{Area}(\triangle OAP) = \frac{1}{2} \cdot 3 \cdot 2 \cdot \sin \frac{\pi}{6} = \frac{3}{2}$$

Hence the area between chord AP and arc AP is

$$\left(\frac{\pi}{4} + \frac{5\sqrt{3}}{8} \right) - \frac{3}{2}$$

Therefore

$$\begin{aligned} \text{Area}(R) &= \frac{3 - \sqrt{3}}{2} - \left[\left(\frac{\pi}{4} + \frac{5\sqrt{3}}{8} \right) - \frac{3}{2} \right] \\ &= \frac{3 - \sqrt{3}}{2} + \frac{3}{2} - \frac{\pi}{4} - \frac{5\sqrt{3}}{8} \\ &= 3 - \frac{\pi}{4} - \frac{9\sqrt{3}}{8} \end{aligned}$$

So the exact area of R is

$$3 - \frac{\pi}{4} - \frac{9\sqrt{3}}{8}$$

equivalently

$$\frac{24 - 2\pi - 9\sqrt{3}}{8}$$

2. The curve C has polar equation

$$r = 2 \sin 3\theta, \quad 0 \leq \theta \leq \frac{\pi}{3}$$

At the non-pole point P on C , the tangent to C is parallel to the initial line.

Given that O is the pole, find the exact length of the line OP .

[7]

Solution

For a polar curve,

$$x = r \cos \theta, \quad y = r \sin \theta, \quad \frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$$

A tangent parallel to the initial line is horizontal, so its gradient is 0. Hence we require

$$\frac{dy}{d\theta} = 0$$

with $\frac{dx}{d\theta} \neq 0$.

Here

$$r = 2 \sin 3\theta$$

so

$$\frac{dr}{d\theta} = 6 \cos 3\theta$$

Now

$$y = r \sin \theta$$

so, by the product rule,

$$\begin{aligned} \frac{dy}{d\theta} &= \frac{dr}{d\theta} \sin \theta + r \cos \theta \\ &= 6 \cos 3\theta \sin \theta + 2 \sin 3\theta \cos \theta \end{aligned}$$

Set this equal to 0:

$$6 \cos 3\theta \sin \theta + 2 \sin 3\theta \cos \theta = 0$$

Divide by 2:

$$3 \cos 3\theta \sin \theta + \sin 3\theta \cos \theta = 0$$

Using the product-to-sum identities,

$$\sin \theta \cos 3\theta = \frac{1}{2}(\sin 4\theta - \sin 2\theta), \quad \cos \theta \sin 3\theta = \frac{1}{2}(\sin 4\theta + \sin 2\theta)$$

So

$$\begin{aligned} 0 &= 3 \cdot \frac{1}{2}(\sin 4\theta - \sin 2\theta) + \frac{1}{2}(\sin 4\theta + \sin 2\theta) \\ &= \frac{1}{2}(4 \sin 4\theta - 2 \sin 2\theta) \end{aligned}$$

Hence

$$2 \sin 4\theta - \sin 2\theta = 0$$

Using $\sin 4\theta = 2 \sin 2\theta \cos 2\theta$,

$$\begin{aligned} 0 &= 2(2 \sin 2\theta \cos 2\theta) - \sin 2\theta \\ &= \sin 2\theta(4 \cos 2\theta - 1) \end{aligned}$$

At a non-pole point on this curve, $0 < \theta < \frac{\pi}{3}$, so $0 < 2\theta < \frac{2\pi}{3}$, and therefore $\sin 2\theta \neq 0$. Thus

$$4 \cos 2\theta - 1 = 0$$

so

$$\cos 2\theta = \frac{1}{4}$$

Now use $\cos 2\theta = 2\cos^2 \theta - 1$:

$$2\cos^2 \theta - 1 = \frac{1}{4}$$

$$2\cos^2 \theta = \frac{5}{4}$$

$$\cos^2 \theta = \frac{5}{8}$$

Therefore

$$\sin^2 \theta = 1 - \frac{5}{8} = \frac{3}{8}$$

Since $0 < \theta < \frac{\pi}{3}$, both $\sin \theta$ and $\cos \theta$ are positive, so

$$\sin \theta = \sqrt{\frac{3}{8}} = \frac{\sqrt{6}}{4}$$

Now

$$r = 2 \sin 3\theta$$

and using $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$,

$$\begin{aligned} r &= 2 \left(3 \cdot \frac{\sqrt{6}}{4} - 4 \left(\frac{\sqrt{6}}{4} \right)^3 \right) \\ &= 2 \left(\frac{3\sqrt{6}}{4} - 4 \cdot \frac{6\sqrt{6}}{64} \right) \\ &= 2 \left(\frac{3\sqrt{6}}{4} - \frac{3\sqrt{6}}{8} \right) \\ &= 2 \left(\frac{3\sqrt{6}}{8} \right) \\ &= \frac{3\sqrt{6}}{4} \end{aligned}$$

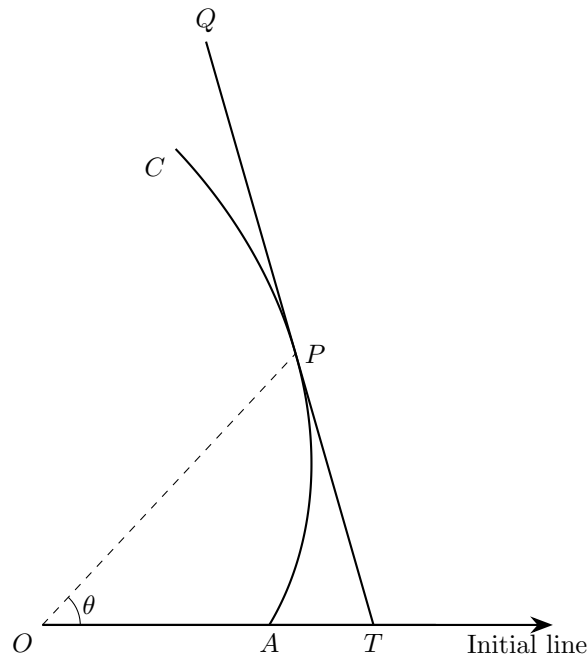
Finally, check that this really is a horizontal tangent:

$$\begin{aligned} \frac{dx}{d\theta} &= \frac{dr}{d\theta} \cos \theta - r \sin \theta \\ &= 6 \cos 3\theta \cos \theta - 2 \sin 3\theta \sin \theta \\ &= -3 \neq 0 \end{aligned}$$

So the tangent is genuinely parallel to the initial line.

Hence the exact length of OP is

$$OP = r = \frac{3\sqrt{6}}{4}$$



3. The diagram shows part of a spiral C .

The point P has polar coordinates (r, θ) , where $0 \leq \theta \leq \frac{\pi}{2}$.

The tangent at P is the line TPQ . The point A lies on C and on the initial line, with $OA = 2$.

Let ψ be the angle that TPQ makes with the initial line.

(a) By expressing the Cartesian coordinates of P in terms of r and θ , show that

$$\tan \psi = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta} \quad [4]$$

(b) The spiral is equiangular: at every point P , the acute angle between OP and TPQ is a constant α , where $0 < \alpha < \frac{\pi}{2}$.

Use the result of part (a) to show that the polar equation of C is

$$r = 2e^{(\cot \alpha)\theta} \quad [7]$$

Solution

(a) Let the Cartesian coordinates of P be (x, y) . Since P has polar coordinates (r, θ) ,

$$x = r \cos \theta, \quad y = r \sin \theta$$

Differentiate both with respect to θ :

$$\begin{aligned} \frac{dx}{d\theta} &= \frac{dr}{d\theta} \cos \theta - r \sin \theta \\ \frac{dy}{d\theta} &= \frac{dr}{d\theta} \sin \theta + r \cos \theta \end{aligned}$$

The angle ψ is the angle the tangent makes with the initial line, so

$$\tan \psi = \frac{dy}{dx}$$

Using

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

gives

$$\tan \psi = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta}$$

So

$$\tan \psi = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta}$$

as required.

(b) Let

$$r' = \frac{dr}{d\theta}$$

From part (a),

$$\tan \psi = \frac{r' \sin \theta + r \cos \theta}{r' \cos \theta - r \sin \theta}$$

Divide top and bottom by $r' \cos \theta$:

$$\tan \psi = \frac{\tan \theta + \frac{r}{r'}}{1 - \tan \theta \frac{r}{r'}}$$

Now let β be defined by

$$\tan \beta = \frac{r}{r'}$$

Then

$$\begin{aligned} \tan \psi &= \frac{\tan \theta + \tan \beta}{1 - \tan \theta \tan \beta} \\ &= \tan(\theta + \beta) \end{aligned}$$

So the angle between OP and the tangent is β .

But the spiral is equiangular, so this acute angle is the constant α . Hence

$$\beta = \alpha$$

and therefore

$$\tan \alpha = \frac{r}{r'}$$

So

$$\begin{aligned} \frac{r}{\frac{dr}{d\theta}} &= \tan \alpha \\ \frac{dr}{d\theta} &= r \cot \alpha \end{aligned}$$

This is a separable differential equation:

$$\frac{1}{r} dr = \cot \alpha d\theta$$

Integrating,

$$\int \frac{1}{r} \, dr = \int \cot \alpha \, d\theta$$
$$\ln r = (\cot \alpha)\theta + c$$

At the point A , the curve is on the initial line and $OA = 2$, so

$$\theta = 0, \quad r = 2$$

Substitute into the equation:

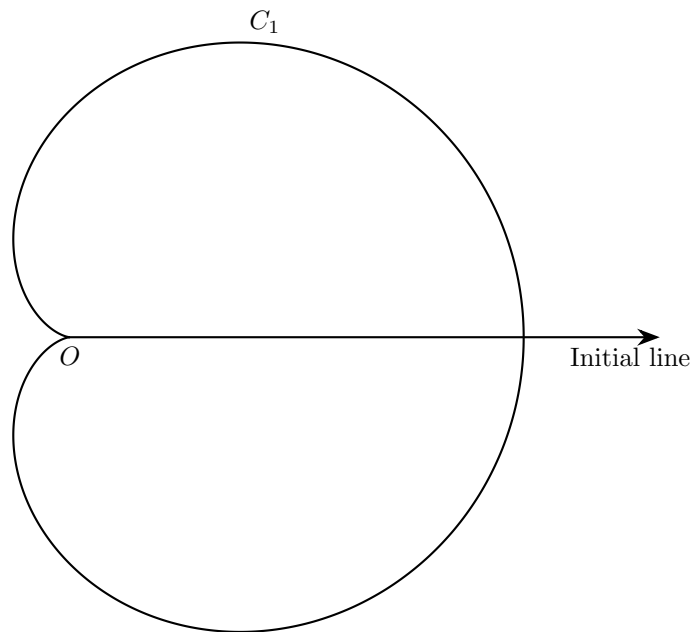
$$\ln 2 = (\cot \alpha)(0) + c$$
$$c = \ln 2$$

Hence

$$\ln r = (\cot \alpha)\theta + \ln 2$$
$$r = 2e^{(\cot \alpha)\theta}$$

Therefore the polar equation of C is

$$r = 2e^{(\cot \alpha)\theta}$$



4. The diagram shows the sketch of a curve C_1 .

The polar equation of the curve C_1 is

$$r = 2(1 + \cos \theta), \quad -\pi \leq \theta \leq \pi$$

(a) Find the area of the region bounded by the curve C_1 . [5]

(b) The curve C_2 whose polar equation is

$$r = \sin \theta, \quad 0 \leq \theta \leq \pi$$

intersects the curve C_1 at the pole O and at the point A . The straight line drawn through A perpendicular to the initial line intersects the part of C_1 above the initial line again at the point B .

(i) Find the polar coordinates of A . [4]

(ii) Show that the length of OB is $\frac{6}{5}$. [6]

(iii) Find the length of AB , giving your answer to three significant figures. [3]

Solution

(a) Using the polar area formula,

$$\text{Area} = \frac{1}{2} \int_{-\pi}^{\pi} r^2 \, d\theta$$

Here $r = 2(1 + \cos \theta)$, so

$$\text{Area} = \frac{1}{2} \int_{-\pi}^{\pi} (2(1 + \cos \theta))^2 \, d\theta = 2 \int_{-\pi}^{\pi} (1 + 2 \cos \theta + \cos^2 \theta) \, d\theta$$

$$\text{Using } \cos^2 \theta = \frac{1 + \cos 2\theta}{2},$$

$$\text{Area} = 2 \int_{-\pi}^{\pi} \left(1 + 2 \cos \theta + \frac{1 + \cos 2\theta}{2} \right) \, d\theta$$

$$\begin{aligned}
&= 2 \int_{-\pi}^{\pi} \left(\frac{3}{2} + 2 \cos \theta + \frac{1}{2} \cos 2\theta \right) d\theta \\
&= 2 \left[\frac{3}{2} \theta + 2 \sin \theta + \frac{1}{4} \sin 2\theta \right]_{-\pi}^{\pi}
\end{aligned}$$

At $\theta = \pi$ and $\theta = -\pi$, the sine terms are 0, so

$$\text{Area} = 2 \left(\frac{3}{2} \pi - \left(-\frac{3}{2} \pi \right) \right) = 2(3\pi) = 6\pi$$

Hence the area of the region bounded by C_1 is 6π .

- (b) (i) Since both curves have $r \geq 0$ on the given ranges, any intersection away from the pole must occur for the same value of θ .

So we solve

$$2(1 + \cos \theta) = \sin \theta$$

Squaring both sides,

$$4(1 + \cos \theta)^2 = \sin^2 \theta$$

Using $\sin^2 \theta = 1 - \cos^2 \theta$,

$$4(1 + 2 \cos \theta + \cos^2 \theta) = 1 - \cos^2 \theta$$

$$4 + 8 \cos \theta + 4 \cos^2 \theta = 1 - \cos^2 \theta$$

$$5 \cos^2 \theta + 8 \cos \theta + 3 = 0$$

$$(5 \cos \theta + 3)(\cos \theta + 1) = 0$$

So

$$\cos \theta = -\frac{3}{5} \quad \text{or} \quad \cos \theta = -1$$

The value $\cos \theta = -1$ gives $\theta = \pi$, and then $r = \sin \pi = 0$, which is the pole O .

Therefore point A corresponds to

$$\cos \theta = -\frac{3}{5}$$

Since $0 \leq \theta \leq \pi$, we have $\sin \theta > 0$, so

$$\sin \theta = \sqrt{1 - \left(-\frac{3}{5}\right)^2} = \sqrt{1 - \frac{9}{25}} = \frac{4}{5}$$

Hence

$$r = \sin \theta = \frac{4}{5}$$

So the polar coordinates of A are

$$\left(\frac{4}{5}, \arccos \left(-\frac{3}{5} \right) \right)$$

Equivalently, $\theta \approx 126.9^\circ$.

- (ii) The line through A perpendicular to the initial line is a vertical line, so points A and B have the same x -coordinate.

For A ,

$$x_A = r \cos \theta = \frac{4}{5} \cdot \left(-\frac{3}{5} \right) = -\frac{12}{25}$$

For a general point on C_1 ,

$$r = 2(1 + \cos \theta)$$

so

$$x = r \cos \theta = 2 \cos \theta (1 + \cos \theta)$$

At point B , $x = x_A$, so

$$2 \cos \theta (1 + \cos \theta) = -\frac{12}{25}$$

Let $c = \cos \theta$. Then

$$2c(1 + c) = -\frac{12}{25}$$

$$50c^2 + 50c = -12$$

$$25c^2 + 25c + 6 = 0$$

$$(5c + 3)(5c + 2) = 0$$

So

$$c = -\frac{3}{5} \quad \text{or} \quad c = -\frac{2}{5}$$

The root $c = -\frac{3}{5}$ corresponds to A , so for B ,

$$\cos \theta = -\frac{2}{5}$$

Hence

$$r_B = 2 \left(1 - \frac{2}{5} \right) = 2 \cdot \frac{3}{5} = \frac{6}{5}$$

Since OB is the distance from the pole to B ,

$$OB = \frac{6}{5}$$

So the length of OB is $\frac{6}{5}$.

(iii) Since AB is vertical, its length is the difference between the y -coordinates of B and A .

For A ,

$$y_A = r \sin \theta = \frac{4}{5} \cdot \frac{4}{5} = \frac{16}{25}$$

For B , we have $\cos \theta = -\frac{2}{5}$, and B is above the initial line, so

$$\sin \theta = \sqrt{1 - \left(-\frac{2}{5} \right)^2} = \sqrt{1 - \frac{4}{25}} = \frac{\sqrt{21}}{5}$$

Therefore

$$y_B = r \sin \theta = \frac{6}{5} \cdot \frac{\sqrt{21}}{5} = \frac{6\sqrt{21}}{25}$$

So

$$AB = y_B - y_A = \frac{6\sqrt{21}}{25} - \frac{16}{25} = \frac{6\sqrt{21} - 16}{25}$$

$$AB \approx 0.460$$

Hence

$$AB = \frac{6\sqrt{21} - 16}{25} \approx 0.460$$

to three significant figures.

5. The curve C has polar equation

$$r = 3 \sin 2\theta, \quad 0 \leq \theta \leq \frac{\pi}{2}$$

At the non-pole point P on C , the tangent to C is perpendicular to the initial line.

Given that O is the pole, find the exact length of the line OP .

[7]

Solution

For a polar curve,

$$x = r \cos \theta, \quad y = r \sin \theta$$

so

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{r' \sin \theta + r \cos \theta}{r' \cos \theta - r \sin \theta}$$

A tangent perpendicular to the initial line is vertical, so $\frac{dy}{dx}$ is undefined. Therefore

$$r' \cos \theta - r \sin \theta = 0$$

with $r' \sin \theta + r \cos \theta \neq 0$

Given

$$r = 3 \sin 2\theta$$

we have

$$r' = 6 \cos 2\theta$$

Substituting into the vertical tangent condition gives

$$6 \cos 2\theta \cos \theta - 3 \sin 2\theta \sin \theta = 0$$

Now use

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta, \quad \sin 2\theta = 2 \sin \theta \cos \theta$$

Then

$$\begin{aligned} 6(\cos^2 \theta - \sin^2 \theta) \cos \theta - 3(2 \sin \theta \cos \theta) \sin \theta &= 0 \\ 6(\cos^2 \theta - \sin^2 \theta) \cos \theta - 6 \sin^2 \theta \cos \theta &= 0 \\ 6 \cos \theta (\cos^2 \theta - 2 \sin^2 \theta) &= 0 \end{aligned}$$

Using $\sin^2 \theta = 1 - \cos^2 \theta$,

$$\begin{aligned} 6 \cos \theta (\cos^2 \theta - 2(1 - \cos^2 \theta)) &= 0 \\ 6 \cos \theta (3 \cos^2 \theta - 2) &= 0 \end{aligned}$$

Since P is a non-pole point and $0 \leq \theta \leq \frac{\pi}{2}$, we must have

$$0 < \theta < \frac{\pi}{2}$$

so $\cos \theta \neq 0$

Hence

$$3 \cos^2 \theta - 2 = 0$$

so

$$\cos^2 \theta = \frac{2}{3}$$

Also

$$\sin^2 \theta = 1 - \frac{2}{3} = \frac{1}{3}$$

Because $0 < \theta < \frac{\pi}{2}$, both sine and cosine are positive, so

$$\cos \theta = \sqrt{\frac{2}{3}}, \quad \sin \theta = \frac{1}{\sqrt{3}}$$

Now

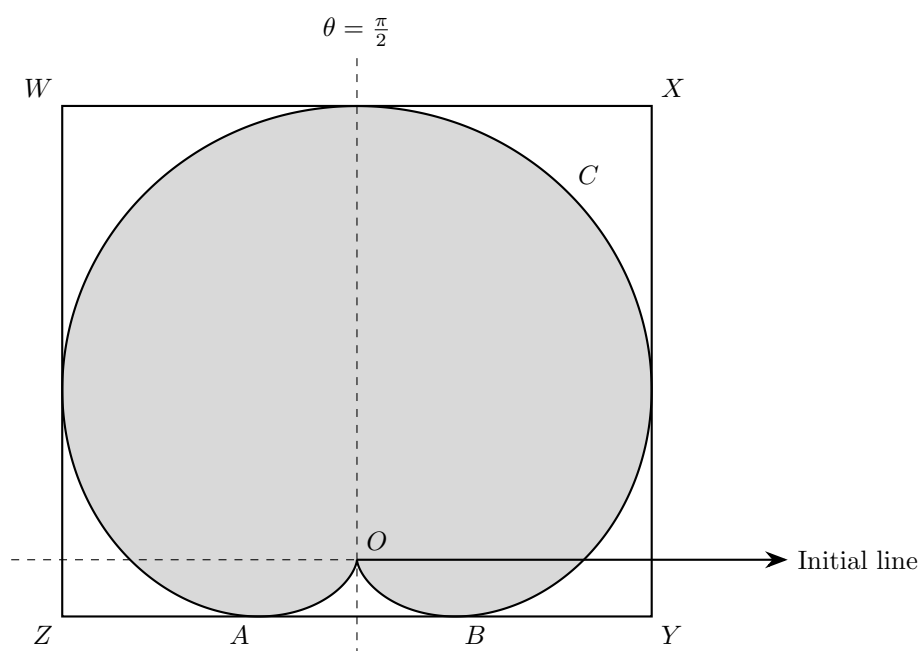
$$\begin{aligned} r &= 3 \sin 2\theta \\ &= 3(2 \sin \theta \cos \theta) \\ &= 6 \cdot \frac{1}{\sqrt{3}} \cdot \sqrt{\frac{2}{3}} \\ &= 2\sqrt{2} \end{aligned}$$

Finally, check that the numerator is non-zero:

$$\begin{aligned} r' \sin \theta + r \cos \theta &= 6 \cos 2\theta \sin \theta + 3 \sin 2\theta \cos \theta \\ &= 6 \left(\frac{1}{3} \right) \left(\frac{1}{\sqrt{3}} \right) + 3 \left(\frac{2\sqrt{2}}{3} \right) \left(\sqrt{\frac{2}{3}} \right) \\ &= \frac{2}{\sqrt{3}} + \frac{4}{\sqrt{3}} \\ &= \frac{6}{\sqrt{3}} = 2\sqrt{3} \neq 0 \end{aligned}$$

So the tangent is genuinely perpendicular to the initial line, and since $OP = r$,

$$OP = 2\sqrt{2}$$



6. The figure shows a sketch of the cardioid C with equation

$$r = a(1 + \sin \theta), \quad 0 \leq \theta < 2\pi$$

Also shown are the tangents to C that are parallel and perpendicular to the initial line. These tangents form a rectangle $WXYZ$.

(a) Find the area of the finite region, shaded in the figure, enclosed by C . [6]

(b) Find the values of θ for which the tangent to C is parallel to the initial line. Hence find the polar coordinates of the points A and B , where ZY touches C . [6]

(c) Hence show that the length of WZ is $\frac{9a}{4}$. [2]

Given that the length of WX is $\frac{3\sqrt{3}a}{2}$

(d) find the area of the rectangle $WXYZ$. [2]

A logo is modelled by the cardioid C , where $a = 8$ cm. The logo is cut from the rectangular card $WXYZ$, shown in the figure.

(e) Find a numerical value for the area of card wasted in making the logo. [3]

Solution

(a) For a polar curve $r = f(\theta)$, the area enclosed is

$$\frac{1}{2} \int r^2 d\theta$$

Here $r = a(1 + \sin \theta)$, and $0 \leq \theta < 2\pi$ traces the whole cardioid once, so

$$\begin{aligned}\text{Area} &= \frac{1}{2} \int_0^{2\pi} a^2 (1 + \sin \theta)^2 d\theta \\ &= \frac{a^2}{2} \int_0^{2\pi} (1 + 2 \sin \theta + \sin^2 \theta) d\theta\end{aligned}$$

Using $\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$,

$$\begin{aligned}\text{Area} &= \frac{a^2}{2} \int_0^{2\pi} \left(1 + 2 \sin \theta + \frac{1 - \cos 2\theta}{2} \right) d\theta \\ &= \frac{a^2}{2} \int_0^{2\pi} \left(\frac{3}{2} + 2 \sin \theta - \frac{1}{2} \cos 2\theta \right) d\theta \\ &= \frac{a^2}{2} \left[\frac{3}{2} \theta - 2 \cos \theta - \frac{1}{4} \sin 2\theta \right]_0^{2\pi} \\ &= \frac{a^2}{2} (3\pi)\end{aligned}$$

Therefore the area enclosed by C is

$$\frac{3\pi a^2}{2}$$

(b) Let

$$x = r \cos \theta, \quad y = r \sin \theta$$

with

$$r = a(1 + \sin \theta), \quad \frac{dr}{d\theta} = a \cos \theta$$

Also,

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

A tangent parallel to the initial line is horizontal, so we need

$$\frac{dy}{d\theta} = 0$$

with $\frac{dx}{d\theta} \neq 0$.

Now

$$\begin{aligned}\frac{dy}{d\theta} &= \frac{d}{d\theta}(r \sin \theta) \\ &= \frac{dr}{d\theta} \sin \theta + r \cos \theta \\ &= a \cos \theta \sin \theta + a(1 + \sin \theta) \cos \theta \\ &= a \cos \theta (1 + 2 \sin \theta)\end{aligned}$$

So

$$a \cos \theta (1 + 2 \sin \theta) = 0$$

Hence either

$$\cos \theta = 0 \quad \text{or} \quad 1 + 2 \sin \theta = 0$$

From $\cos \theta = 0$,

$$\theta = \frac{\pi}{2}, \quad \frac{3\pi}{2}$$

From $1 + 2 \sin \theta = 0$,

$$\sin \theta = -\frac{1}{2}$$

so

$$\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$$

At $\theta = \frac{3\pi}{2}$,

$$r = a(1 + \sin \frac{3\pi}{2}) = a(1 - 1) = 0$$

so this is the cusp at the pole, not a smooth horizontal tangent.

Therefore the values of θ for which the tangent is parallel to the initial line are

$$\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

The line ZY is the lower horizontal tangent, so it touches C at the two lower points, where

$$\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$$

At each of these,

$$r = a(1 + \sin \theta) = a\left(1 - \frac{1}{2}\right) = \frac{a}{2}$$

Hence the polar coordinates are

$$A\left(\frac{a}{2}, \frac{7\pi}{6}\right), \quad B\left(\frac{a}{2}, \frac{11\pi}{6}\right)$$

(c) From part (b), the upper horizontal tangent occurs at $\theta = \frac{\pi}{2}$.

Then

$$r = a(1 + \sin \frac{\pi}{2}) = 2a$$

so its y -coordinate is

$$y = r \sin \theta = 2a \cdot 1 = 2a$$

The lower horizontal tangent touches at A and B , where $r = \frac{a}{2}$ and $\theta = \frac{7\pi}{6}$ or $\frac{11\pi}{6}$, so

$$y = \frac{a}{2} \sin \frac{7\pi}{6} = \frac{a}{2} \left(-\frac{1}{2}\right) = -\frac{a}{4}$$

Therefore the vertical distance between these tangents is

$$WZ = 2a - \left(-\frac{a}{4}\right) = \frac{9a}{4}$$

So

$$WZ = \frac{9a}{4}$$

(d) The area of the rectangle is

$$\text{Area} = WX \times WZ$$

Given

$$WX = \frac{3\sqrt{3}a}{2} \quad \text{and} \quad WZ = \frac{9a}{4}$$

we get

$$\begin{aligned} \text{Area} &= \frac{3\sqrt{3}a}{2} \times \frac{9a}{4} \\ &= \frac{27\sqrt{3}a^2}{8} \end{aligned}$$

Therefore the area of rectangle $WXYZ$ is

$$\frac{27\sqrt{3}a^2}{8}$$

(e) When $a = 8$,

$$\begin{aligned}\text{Area of rectangle} &= \frac{27\sqrt{3}a^2}{8} \\ &= \frac{27\sqrt{3} \cdot 8^2}{8} \\ &= 216\sqrt{3} \text{ cm}^2\end{aligned}$$

From part (a),

$$\begin{aligned}\text{Area of cardioid} &= \frac{3\pi a^2}{2} \\ &= \frac{3\pi \cdot 8^2}{2} \\ &= 96\pi \text{ cm}^2\end{aligned}$$

So the wasted area is

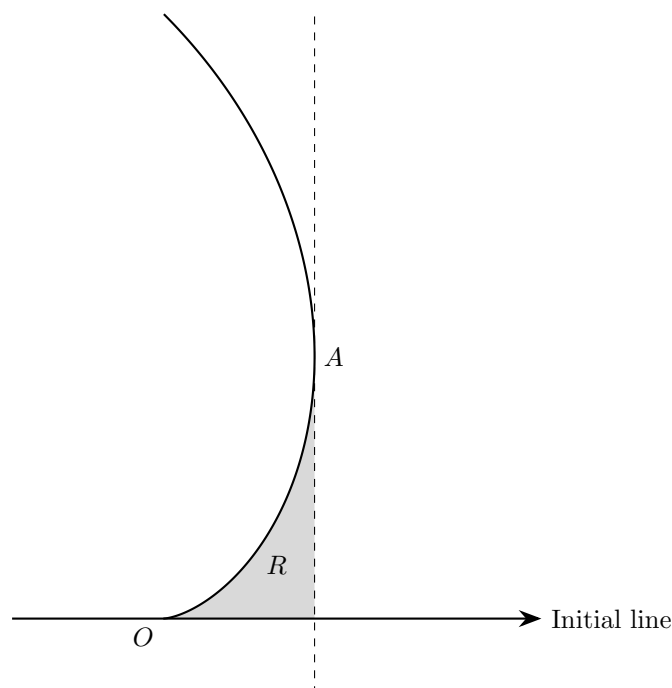
$$216\sqrt{3} - 96\pi \text{ cm}^2$$

Numerically,

$$216\sqrt{3} - 96\pi \approx 72.5$$

Therefore the area of card wasted is

$$216\sqrt{3} - 96\pi \text{ cm}^2 \approx 72.5 \text{ cm}^2$$



7. The diagram above shows a sketch of the curve C with equation

$$r = 4(1 - \cos \theta) \quad 0 \leq \theta \leq \frac{\pi}{2}$$

The diagram above also shows the tangent to C at the point A , where A is not the pole. This tangent is perpendicular to the initial line.

(a) Use differentiation to prove that the polar coordinates of A are $\left(2, \frac{\pi}{3}\right)$. [4]

The finite region R , shown shaded in the diagram above, is bounded by C , the tangent at A and the initial line.

(b) Use calculus to show that the exact area of R is $\frac{1}{2}(15\sqrt{3} - 8\pi)$. [6]

Solution

(a) For a polar curve,

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

Here

$$x = r \cos \theta = 4(1 - \cos \theta) \cos \theta, \quad y = r \sin \theta = 4(1 - \cos \theta) \sin \theta$$

The initial line is the positive x -axis, so a tangent perpendicular to the initial line is a vertical tangent. Therefore we need

$$\frac{dx}{d\theta} = 0 \quad \text{and} \quad \frac{dy}{d\theta} \neq 0$$

Now

$$\begin{aligned}x &= 4(\cos \theta - \cos^2 \theta) \\ \frac{dx}{d\theta} &= 4(-\sin \theta + 2 \sin \theta \cos \theta) \\ &= 4 \sin \theta (2 \cos \theta - 1)\end{aligned}$$

So

$$4 \sin \theta (2 \cos \theta - 1) = 0$$

Hence

$$\sin \theta = 0 \quad \text{or} \quad 2 \cos \theta - 1 = 0$$

Since $0 \leq \theta \leq \frac{\pi}{2}$, this gives

$$\theta = 0 \quad \text{or} \quad \cos \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$$

At $\theta = 0$,

$$r = 4(1 - \cos 0) = 4(1 - 1) = 0$$

so this is the pole, but A is not the pole.

Therefore

$$\theta = \frac{\pi}{3}$$

Now find r :

$$r = 4 \left(1 - \cos \frac{\pi}{3} \right) = 4 \left(1 - \frac{1}{2} \right) = 2$$

To confirm the tangent is vertical, differentiate y :

$$\begin{aligned}y &= 4(1 - \cos \theta) \sin \theta \\ \frac{dy}{d\theta} &= 4 (\sin^2 \theta + (1 - \cos \theta) \cos \theta)\end{aligned}$$

At $\theta = \frac{\pi}{3}$,

$$\begin{aligned}\frac{dy}{d\theta} &= 4 \left(\frac{3}{4} + \frac{1}{2} \cdot \frac{1}{2} \right) \\ &= 4 \left(\frac{3}{4} + \frac{1}{4} \right) = 4 \neq 0\end{aligned}$$

So the tangent is indeed perpendicular to the initial line there.

Therefore the polar coordinates of A are

$$\left(2, \frac{\pi}{3} \right)$$

(b) Since $A = \left(2, \frac{\pi}{3} \right)$, its Cartesian x -coordinate is

$$x_A = r \cos \theta = 2 \cos \frac{\pi}{3} = 1$$

So the tangent at A is the vertical line $x = 1$.

Hence the shaded region R is the area under the curve from O to A , above the initial line, so

$$A_R = \int y \, dx = \int_0^{\pi/3} y \frac{dx}{d\theta} \, d\theta$$

Using

$$y = 4(1 - \cos \theta) \sin \theta, \quad \frac{dx}{d\theta} = 4 \sin \theta (2 \cos \theta - 1)$$

gives

$$A_R = 16 \int_0^{\pi/3} (1 - \cos \theta) \sin^2 \theta (2 \cos \theta - 1) d\theta$$

Now use $\sin^2 \theta = 1 - \cos^2 \theta$:

$$\begin{aligned} (1 - \cos \theta) \sin^2 \theta (2 \cos \theta - 1) &= (1 - \cos \theta)(1 - \cos^2 \theta)(2 \cos \theta - 1) \\ &= (1 - \cos \theta - \cos^2 \theta + \cos^3 \theta)(2 \cos \theta - 1) \\ &= 2 \cos^4 \theta - 3 \cos^3 \theta - \cos^2 \theta + 3 \cos \theta - 1 \end{aligned}$$

Therefore

$$A_R = 16 \int_0^{\pi/3} (2 \cos^4 \theta - 3 \cos^3 \theta - \cos^2 \theta + 3 \cos \theta - 1) d\theta$$

Use

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

so

$$\begin{aligned} \cos^4 \theta &= \left(\frac{1 + \cos 2\theta}{2} \right)^2 \\ &= \frac{1}{4} (1 + 2 \cos 2\theta + \cos^2 2\theta) \\ &= \frac{1}{4} \left(1 + 2 \cos 2\theta + \frac{1 + \cos 4\theta}{2} \right) \\ &= \frac{1}{8} (3 + 4 \cos 2\theta + \cos 4\theta) \end{aligned}$$

Also,

$$\int \cos^3 \theta d\theta = \int \cos \theta (1 - \sin^2 \theta) d\theta = \sin \theta - \frac{1}{3} \sin^3 \theta$$

So

$$\begin{aligned} A_R &= 16 \left[\int \left(\frac{3}{4} + \cos 2\theta + \frac{1}{4} \cos 4\theta \right) d\theta - 3 \left(\sin \theta - \frac{1}{3} \sin^3 \theta \right) \right. \\ &\quad \left. - \left(\frac{1}{2} \theta + \frac{1}{4} \sin 2\theta \right) + 3 \sin \theta - \theta \right]_0^{\pi/3} \\ &= 16 \left[\frac{3}{4} \theta + \frac{1}{2} \sin 2\theta + \frac{1}{16} \sin 4\theta - 3 \sin \theta + \sin^3 \theta - \frac{1}{2} \theta - \frac{1}{4} \sin 2\theta + 3 \sin \theta - \theta \right]_0^{\pi/3} \\ &= 16 \left[\sin^3 \theta + \frac{1}{4} \sin 2\theta + \frac{1}{16} \sin 4\theta - \frac{3}{4} \theta \right]_0^{\pi/3} \end{aligned}$$

Now substitute the limits:

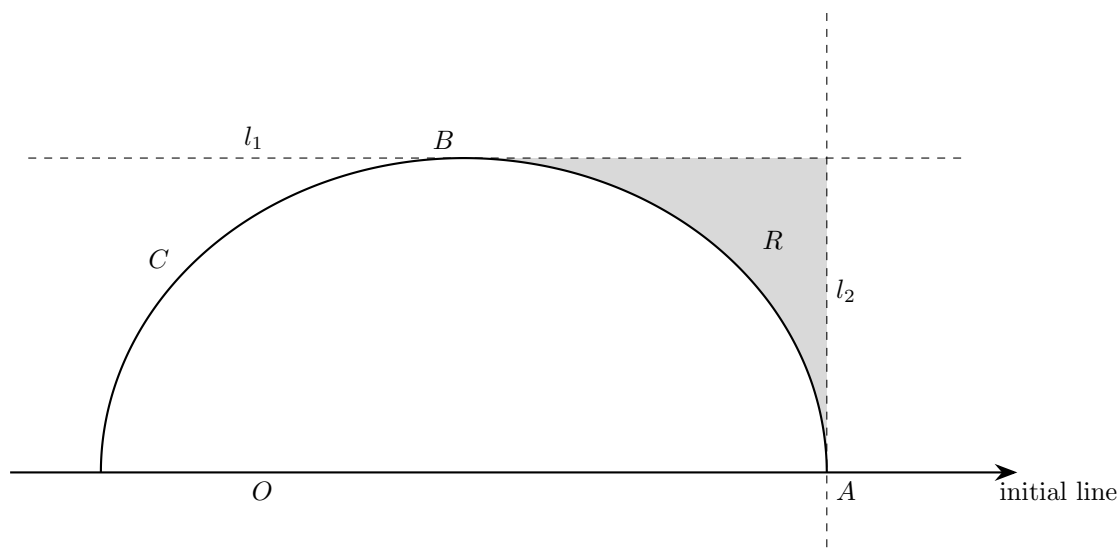
$$\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}, \quad \sin \frac{2\pi}{3} = \frac{\sqrt{3}}{2}, \quad \sin \frac{4\pi}{3} = -\frac{\sqrt{3}}{2}$$

Hence

$$\begin{aligned}A_R &= 16 \left[\left(\frac{\sqrt{3}}{2} \right)^3 + \frac{1}{4} \cdot \frac{\sqrt{3}}{2} + \frac{1}{16} \cdot \left(-\frac{\sqrt{3}}{2} \right) - \frac{3}{4} \cdot \frac{\pi}{3} \right] \\&= 16 \left[\frac{3\sqrt{3}}{8} + \frac{\sqrt{3}}{8} - \frac{\sqrt{3}}{32} - \frac{\pi}{4} \right] \\&= 16 \left[\frac{15\sqrt{3}}{32} - \frac{\pi}{4} \right] \\&= \frac{15\sqrt{3}}{2} - 4\pi \\&= \frac{1}{2}(15\sqrt{3} - 8\pi)\end{aligned}$$

Therefore the exact area of R is

$$\frac{1}{2}(15\sqrt{3} - 8\pi)$$



8. The figure above shows a sketch of the curve C with equation

$$r = \frac{6}{2 - \cos \theta} \quad 0 \leq \theta \leq \pi$$

Given that C meets the initial line at the point A , as shown in the figure above

- (a) write down the polar coordinates of A . [1]

The line ℓ_1 , also shown in the figure above, is the tangent to C at the point B and is parallel to the initial line.

- (b) Use calculus to determine the polar coordinates of B . [4]

The line ℓ_2 , also shown in the figure above, is the tangent to C at A and is perpendicular to the initial line.

The region R , shown shaded in the figure above, is bounded by C , ℓ_1 and ℓ_2 .

- (c) By first obtaining a Cartesian equation for C , use algebraic integration to find the exact area of R , giving your answer in the form $\sqrt{3}(p + q\pi)$ where p and q are constants to be determined. [8]

Solution

- (a) On the initial line, $\theta = 0$.

$$r = \frac{6}{2 - \cos 0} = \frac{6}{2 - 1} = 6$$

So the polar coordinates of A are $(6, 0)$.

- (b) Using

$$x = r \cos \theta, \quad y = r \sin \theta$$

we have

$$x = \frac{6 \cos \theta}{2 - \cos \theta}, \quad y = \frac{6 \sin \theta}{2 - \cos \theta}$$

For a tangent parallel to the initial line, we need

$$\frac{dy}{dx} = 0$$

and

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

Now differentiate x and y with respect to θ .

$$\begin{aligned}\frac{dx}{d\theta} &= \frac{(-6 \sin \theta)(2 - \cos \theta) - 6 \cos \theta(\sin \theta)}{(2 - \cos \theta)^2} \\ &= \frac{-12 \sin \theta + 6 \sin \theta \cos \theta - 6 \sin \theta \cos \theta}{(2 - \cos \theta)^2} \\ &= \frac{-12 \sin \theta}{(2 - \cos \theta)^2}\end{aligned}$$

For $0 < \theta < \pi$, $\sin \theta \neq 0$, so $\frac{dx}{d\theta} \neq 0$. Hence a horizontal tangent occurs when

$$\frac{dy}{d\theta} = 0$$

Now

$$\begin{aligned}\frac{dy}{d\theta} &= \frac{6 \cos \theta(2 - \cos \theta) - 6 \sin^2 \theta}{(2 - \cos \theta)^2} \\ &= \frac{12 \cos \theta - 6 \cos^2 \theta - 6 \sin^2 \theta}{(2 - \cos \theta)^2} \\ &= \frac{12 \cos \theta - 6(\cos^2 \theta + \sin^2 \theta)}{(2 - \cos \theta)^2} \\ &= \frac{12 \cos \theta - 6}{(2 - \cos \theta)^2} \\ &= \frac{6(2 \cos \theta - 1)}{(2 - \cos \theta)^2}\end{aligned}$$

So

$$\frac{dy}{d\theta} = 0 \quad \Rightarrow \quad 2 \cos \theta - 1 = 0 \quad \Rightarrow \quad \cos \theta = \frac{1}{2}$$

Since $0 \leq \theta \leq \pi$,

$$\theta = \frac{\pi}{3}$$

Then

$$r = \frac{6}{2 - \cos\left(\frac{\pi}{3}\right)} = \frac{6}{2 - \frac{1}{2}} = \frac{6}{\frac{3}{2}} = 4$$

So the polar coordinates of B are

$$\left(4, \frac{\pi}{3}\right)$$

(c) Start with the polar equation

$$r = \frac{6}{2 - \cos \theta}$$

Rearranging,

$$r(2 - \cos \theta) = 6$$

Using $x = r \cos \theta$, this gives

$$2r - x = 6$$

Also, since $r^2 = x^2 + y^2$,

$$2r = x + 6$$

Squaring both sides,

$$\begin{aligned} 4r^2 &= (x + 6)^2 \\ 4(x^2 + y^2) &= x^2 + 12x + 36 \\ 3x^2 + 4y^2 - 12x - 36 &= 0 \end{aligned}$$

Complete the square in x :

$$\begin{aligned} 3(x^2 - 4x) + 4y^2 &= 36 \\ 3((x - 2)^2 - 4) + 4y^2 &= 36 \\ 3(x - 2)^2 + 4y^2 &= 48 \end{aligned}$$

So a Cartesian equation for C is

$$3(x - 2)^2 + 4y^2 = 48$$

Since $0 \leq \theta \leq \pi$, the curve lies above the x -axis, so

$$y = \frac{\sqrt{3}}{2} \sqrt{16 - (x - 2)^2}$$

Now find the bounding lines.

At $A = (6, 0)$, the tangent l_2 is perpendicular to the initial line, so it is vertical:

$$l_2 : x = 6$$

From part (b), $B = (4, \pi/3)$, so in Cartesian coordinates

$$x = 4 \cos \frac{\pi}{3} = 2, \quad y = 4 \sin \frac{\pi}{3} = 2\sqrt{3}$$

Since l_1 is parallel to the initial line, it is horizontal:

$$l_1 : y = 2\sqrt{3}$$

Hence the area of R is

$$\int_2^6 \left(2\sqrt{3} - \frac{\sqrt{3}}{2} \sqrt{16 - (x - 2)^2} \right) dx$$

So

$$\begin{aligned} \text{Area}(R) &= \int_2^6 2\sqrt{3} dx - \frac{\sqrt{3}}{2} \int_2^6 \sqrt{16 - (x - 2)^2} dx \\ &= 8\sqrt{3} - \frac{\sqrt{3}}{2} \int_2^6 \sqrt{16 - (x - 2)^2} dx \end{aligned}$$

Let

$$u = x - 2$$

Then when $x = 2$, $u = 0$, and when $x = 6$, $u = 4$. So

$$\text{Area}(R) = 8\sqrt{3} - \frac{\sqrt{3}}{2} \int_0^4 \sqrt{16 - u^2} du$$

Now use $u = 4 \sin t$. Then

$$du = 4 \cos t \, dt$$

and the limits become $t = 0$ to $t = \frac{\pi}{2}$.

$$\begin{aligned} \int_0^4 \sqrt{16 - u^2} \, du &= \int_0^{\pi/2} \sqrt{16 - 16 \sin^2 t} (4 \cos t) \, dt \\ &= \int_0^{\pi/2} 4 \cos t \cdot 4 \cos t \, dt \\ &= 16 \int_0^{\pi/2} \cos^2 t \, dt \\ &= 16 \int_0^{\pi/2} \frac{1 + \cos 2t}{2} \, dt \\ &= 8 \int_0^{\pi/2} (1 + \cos 2t) \, dt \\ &= 8 \left[t + \frac{1}{2} \sin 2t \right]_0^{\pi/2} \\ &= 8 \left(\frac{\pi}{2} \right) \\ &= 4\pi \end{aligned}$$

Therefore

$$\begin{aligned} \text{Area}(R) &= 8\sqrt{3} - \frac{\sqrt{3}}{2}(4\pi) \\ &= 8\sqrt{3} - 2\pi\sqrt{3} \\ &= \sqrt{3}(8 - 2\pi) \end{aligned}$$

So the exact area is

$$\sqrt{3}(8 - 2\pi)$$

and therefore

$$p = 8, \quad q = -2$$